
$\Delta b_{22}(\kappa_2)$, band κ_2/κ_c :	0.5–0.75	0.75–1.2	1.2–1.9	1.9–3	3–4.8	4.8–7.6	7.6–12
exact projected RDT	+0.146	+0.008	−0.020	−0.041	−0.050	−0.054	−0.056
DNS	+0.032	−0.033	−0.029	−0.032	−0.036	−0.040	−0.046
SC-ODT	+0.103	+0.103	+0.096	+0.073	+0.075	+0.079	+0.076

Table 1. Strain-induced upwash anisotropy per wavenumber band at $e = 1$, $Sk_t/\varepsilon = 0.8$, each system referenced to its own unstrained state. The band below $0.75 \kappa_c$ rests on a few box modes so those results are indicative only.

605 than from the synthetic initial condition. All ODT statistics in this subsection are medians
606 over 1024 realisations per case with bootstrap confidence intervals; median statistics are
607 used because the strained band energies are strongly intermittent across realisations, and
608 ensemble means of spectral ratios are dominated by rare events at any practical sample
609 size.

610 4.6.1. Three-way spectral anisotropy: SC-ODT, exact RDT, DNS

611 The comparison quantity is the strain-induced change of the spectral component anisotropy,
612 $\Delta b_{nn}(\kappa_2) = \phi_{nn}/\sum_m \phi_{mm} - \frac{1}{3}$ at strain e , minus the same quantity evaluated in the system's
613 own unstrained state with each mode mapped to its strained wavenumber. Referencing each
614 system to its own $e = 0$ state cancels viscous decay, which is component-blind, and the
615 longitudinal–transverse structure of the isotropic line spectra, which differs between a
616 three-dimensional box and the component-symmetric ODT line. Figure 12 and Table 1
617 give the result at $e = 1$, $Sk_t/\varepsilon = 0.8$, on common bands of $\kappa_2(e)/\kappa_c(0)$ with κ_c the initial
618 spectral centroid.

619 The structure of the tabulated results is the finding. Exact linear theory concentrates
620 the strain-induced upwash anisotropy at the energy-containing scales and *removes* it from
621 the small scales, because modes aligned with the line are amplified least; the DNS shows
622 the same shape, weakened at the large scales by the nonlinear return—that nonlinear
623 effects absent from linear theory shape the strained anisotropy across scales is likewise the
624 finding of the 4096³ axisymmetric-contraction simulations of Clay & Yeung (2016). SC-
625 ODT distributes its moment-level anisotropy essentially uniformly over wavenumber—the
626 signature, anticipated in § 4.2, of a wavenumber-uniform redistribution operator acting on
627 a uniformly dilated line. The integrated anisotropy is nevertheless correct (the model tracks
628 the closure at the moment level, § 3.4), so the deficiency is one of allocation across scales,
629 not of amount. The transverse splitting $\phi_{11}/\phi_{33} - 1$, a combination of the two transverse
630 components just shown, carries the same reading (not plotted): exact RDT deepens it
631 with wavenumber and the DNS relaxes it towards zero at the small scales, whereas SC-
632 ODT holds it essentially constant—the wavenumber-uniform signature again, now in the
633 neutral-to-streamwise balance rather than the upwash. For the leading-edge application the
634 operative consequence is that the high-wavenumber anisotropy of the upwash spectrum,
635 which the Amiet response weights, has the wrong sign in SC-ODT; the closure element
636 that must carry the correction—a wavenumber-dependent rapid operator in place of the
637 uniform one—is exactly the spectral kernel formulation of § 5, whose introduction ~~exercise~~
638 is thereby motivated by a measured deficit rather than by its association with an identified
639 physical process.

640 4.6.2. Distance from linear theory as a function of strain rapidity

641 The three-way comparison resolves the anisotropy by wavenumber at one strain-rate
642 parameter. A complementary scalar measure tracks the *whole* spectral shape across strain

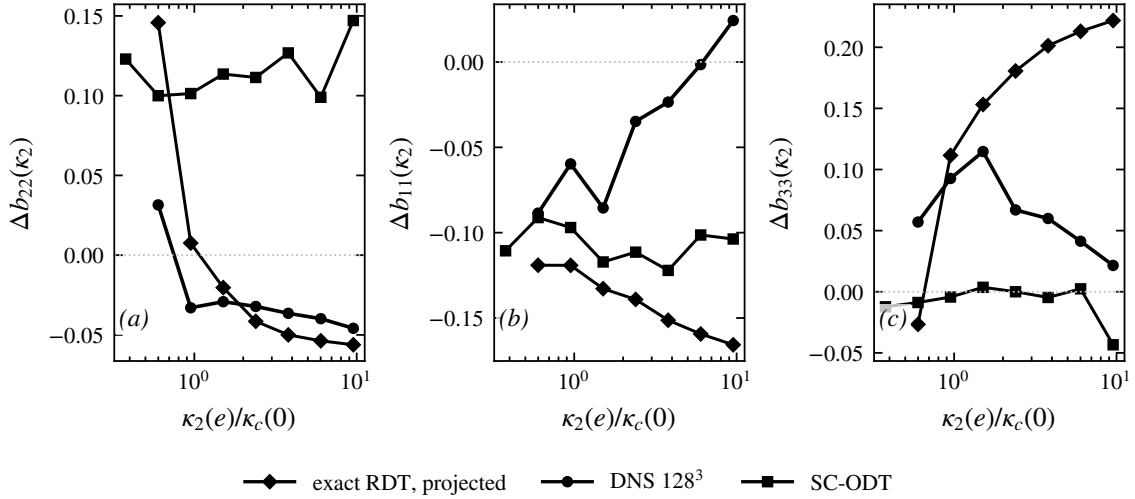


Figure 12. Three-way comparison at $e = 1$, $Sk_t/\varepsilon = 0.8$: strain-induced anisotropy of the line spectra relative to each system's own unstrained state, resolved by component—(a) Δb_{22} , (b) Δb_{11} , (c) Δb_{33} . SC-ODT (■, 1024 realisations) against exact projected RDT (◆) and DNS (128^3 , ●, four realisations). The SC-ODT curves are *near-flat* in every panel—the wavenumber-uniform signature—whereas exact RDT and the DNS vary with wavenumber.

and strain rapidity: at each total strain the median ODT component line spectra are fitted, jointly in the upwash and transverse components, by the exactly distorted spectrum of linear theory—the closed-form Cauchy distortion of an isotropic von Kármán field, projected onto the line, with only the pre-distortion amplitude and energy wavenumber free. The root-mean-square log-residual of that two-parameter fit is a scale-resolved distance between the model's strained spectra and exact linear theory. Figure 13 shows it for precursor-conditioned ensembles at $Sk_t/\varepsilon \approx 0.4$ and ≈ 8 at onset (1024 realisations each, identical precursor state: the two fits coincide at $e = 0$ at thirteen per cent, the residual shape mismatch of a relaxed ODT line against the von Kármán form).

The distance grows with strain in both cases, but *faster at rapid strain*: to approximately 22 per cent at $e = 2$ for $Sk_t/\varepsilon \approx 0.4$ against approximately 24 per cent, saturating early, at $Sk_t/\varepsilon \approx 8$, with the separation established already by $e = 0.5$. If the model's rapid response were spectrally faithful, the rapid curve is the one that should approach zero: eddy events are subdominant there by construction. That the deviation is largest precisely where the eddies matter least localises the deficiency in the rapid kinematics themselves—the wavenumber-uniform operator and the rigid dilatation—and not in the eddy or relaxation statistics, in quantitative agreement with the three-way comparison. This measured validity envelope accompanies the model into any application: predictions inherit a spectral-shape uncertainty of order the plotted distance at the operating point's strain and rapidity.

5. Closure of the rapid pressure–strain term on a single line

The results of § 4 establish what SC-ODT produces and where it stands against exact linear theory and DNS. The rapid pressure–strain redistribution that drives that evolution was evaluated throughout by the algebraic moment closure of § 2 (Launder *et al.* 1975), applied component by component on the resolved line. That closure is a modelling input, and the redistribution it represents is, in its exact form, a *nonlocal, three-dimensional* object: the rapid pressure at a point is a volume integral over the entire fluctuating field, and the pressure–strain correlation is an integral over the full three-dimensional spectrum—one that can be evaluated scale by scale only where the whole field is available, as Clay & Yeung

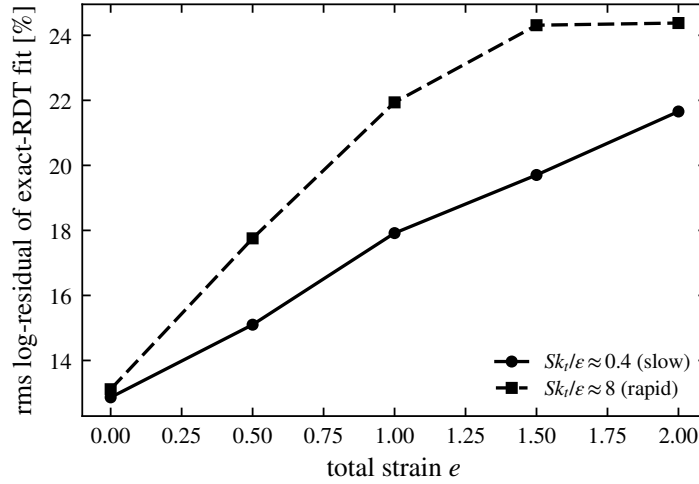


Figure 13. Scale-resolved distance between the strained SC-ODT line spectra and exact linear theory: rms log-residual of the two-parameter exact-RDT-distorted von Kármán fit, against total strain, for slow and rapid onset strain-rate parameters (medians over 1024 realisations; identical precursor state at $e = 0$).

Maybe low and high to avoid apparent association with slow and rapid pressure

(2016) do in a 4096³ simulation of axisymmetric contraction, separating its rapid and slow parts spectrally. What a single one-dimensional line of turbulence can say about such an object is the analytical question of this paper, and this section answers it in a form that is deliberately more modest, and more defensible, than a closure claim: we state exactly what obstructs a single-line representation, we derive sharp bounds on the rapid term from line data alone, we *measure* rather than assume the axisymmetry hypothesis under which the term collapses, and we identify—by measurement—which physical content the moment closure discards.

The section proceeds as follows. The exact term and the obstruction are stated first (§ 5.1–§ 5.2); the axisymmetric collapse follows, together with the statement of what it does *not* deliver (§ 5.3); the bounds that line data do deliver, with their falsifiers, come next (§ 5.4); the axisymmetry assumption is then measured directly in the strained-box DNS, including a control experiment that isolates its failure mode (§ 5.5); the component-ordering deficiency raised by the exact solution is measured and localised (§ 5.6); the reduction to the moment closure and the disjointness from the eddy redistribution close the argument (§ 5.7–§ 5.8).

5.1. The exact rapid pressure–strain term

The starting point is the Poisson equation (2.6) for the fluctuating pressure, whose source splits into a part linear in the fluctuations and the mean velocity gradient (the *rapid* part) and a part quadratic in the fluctuations (the *slow* part) (Pope 2000). The rapid part is linear in $\mathbf{u}'$ and proportional to the mean gradient, which for the present homogeneous plane strain is the constant tensor A_{kl} . Writing p'_r for the rapid pressure and inverting the Laplacian with the free-space Green's function $G(\mathbf{x}) = -1/4\pi|\mathbf{x}|$,

$$p'_r(\mathbf{x}) = \frac{\rho}{2\pi} A_{kl} \int \frac{1}{|\mathbf{x} - \mathbf{x}'|} \frac{\partial u'_l}{\partial x'_k}(\mathbf{x}') d\mathbf{x}'. \quad (5.1)$$

The rapid pressure–strain correlation that redistributes energy among the Reynolds-stress components is $\Pi_{ij}^{(r)} = \overline{(p'_r/\rho) (\partial u'_i/\partial x_j + \partial u'_j/\partial x_i)}$. Substituting (5.1) and using homogeneity to pass to the velocity spectrum tensor $\Phi_{mn}(\boldsymbol{\kappa})$ (the Fourier transform of $\overline{u'_m(\mathbf{x}) u'_n(\mathbf{x} + \mathbf{r})}$), one obtains the standard exact result (Batchelor & Proudman 1954;

Crow 1968; Hunt & Carruthers 1990)

$$\Pi_{ij}^{(r)} = A_{kl} M_{ijkl}, \quad M_{ijkl} = \int_{\mathbb{R}^3} \left[\frac{\kappa_j \kappa_k}{\kappa^2} \Phi_{il}(\boldsymbol{\kappa}) + \frac{\kappa_i \kappa_k}{\kappa^2} \Phi_{jl}(\boldsymbol{\kappa}) \right] d\boldsymbol{\kappa}, \quad (5.2)$$

where $\kappa = |\boldsymbol{\kappa}|$. Equations (2.6)–(5.2) are exact; no modelling has entered. The tensor M_{ijkl} is a linear functional of the *full three-dimensional* spectrum $\Phi_{mn}(\boldsymbol{\kappa})$ through the nonlocal projection operator $\kappa_a \kappa_b / \kappa^2$, which is the wavenumber-space image of the inverse Laplacian in (5.1). This nonlocality and three-dimensionality are the obstruction that the single-line model must confront.

5.2. The obstruction: what an ODT line resolves

ODT evolves a velocity vector $\mathbf{u}'(y)$ on a single line aligned with the resolved direction, here taken along the contraction axis of the strain, $y \equiv x_2$. From it the model has access to the one-dimensional spectra

$$\phi_{mn}(\kappa_2) = \int_{\mathbb{R}^2} \Phi_{mn}(\boldsymbol{\kappa}) d\kappa_1 d\kappa_3, \quad (5.3)$$

the one-dimensional spectra obtained by integrating Φ_{mn} over the two unresolved wavenumber components. The integrand of (5.2), by contrast, carries the *direction-resolved* weight $\kappa_a \kappa_b / \kappa^2$, which cannot be reconstructed from $\phi_{mn}(\kappa_2)$ alone: integrating out κ_1, κ_3 as in (5.3) destroys exactly the angular information that the projection operator acts on. The reduction $M_{ijkl} \rightarrow$ (functional of ϕ_{mn}) is therefore not available without an assumption about how the energy is distributed over the unresolved wavevector directions. This is the crux of the present question, and we make the required assumption explicit rather than implicit.

The obstruction just stated does not disappear under any symmetry assumption short of full isotropy. As § 5.3 shows, axisymmetry about the line reduces the unknown spectrum to two scalar functions of $(\kappa_2, \kappa_\perp)$ —but the line data are two weighted $\kappa_\perp$ -integrals of those functions per resolved wavenumber, so the map from what the line carries to what the rapid term requires retains an infinite-dimensional null space. Stating this null space exactly, and extracting what survives it, is the correct formulation of the single-line question.

5.3. The closure assumption and the collapsed form

The spectrum tensor of the incoming turbulence, before the rapid distortion acts, is axisymmetric about the resolved direction $\mathbf{e}_2$: $\Phi_{mn}(\boldsymbol{\kappa})$ depends on $\boldsymbol{\kappa}$ only through κ_2 and $\kappa_\perp = (\kappa_1^2 + \kappa_3^2)^{1/2}$, and its polarisation is invariant under rotation about $\mathbf{e}_2$.

This is a statement about the *turbulence*, not about the strain. It is justified for the present configuration because the inflow is grid turbulence, which is closely isotropic at the grid and is subsequently contracted along a single axis as it approaches the stagnation plane; isotropy is the special case of A1, and uniaxial contraction preserves axisymmetry about the contraction axis. The plane strain $A_{kl} = \text{diag}(\frac{1}{2}, -\frac{1}{2}, 0)$ acting at the leading edge is itself *not* axisymmetric, so A1 is imposed on the state on which the rapid operator acts, consistent with the rapid-distortion ordering in which $\Pi^{(r)}$ is evaluated on the current (here, axisymmetric to leading order) field. The error incurred when the accumulated distortion has driven the spectrum away from axisymmetry is quantified in § 5.5.

Under A1 the angular integral in (5.2) can be carried out. Writing $\boldsymbol{\kappa} = (\kappa_\perp \cos \theta, \kappa_2, \kappa_\perp \sin \theta)$ and using that, by A1, Φ_{mn} is independent of θ , the azimuthal averages of the projection

weights reduce to

$$\left\langle \frac{\kappa_1^2}{\kappa^2} \right\rangle_\theta = \left\langle \frac{\kappa_3^2}{\kappa^2} \right\rangle_\theta = \frac{1}{2} \frac{\kappa_\perp^2}{\kappa^2}, \quad \left\langle \frac{\kappa_2^2}{\kappa^2} \right\rangle_\theta = \frac{\kappa_2^2}{\kappa^2}, \quad \left\langle \frac{\kappa_1 \kappa_3}{\kappa^2} \right\rangle_\theta = 0, \quad (5.4)$$

with $\kappa^2 = \kappa_2^2 + \kappa_\perp^2$, and all weights mixing a resolved and an unresolved index (e.g. $\kappa_1 \kappa_2 / \kappa^2$) averaging to zero. Substituting (5.4) into (5.2) removes the azimuthal dependence and leaves a two-dimensional integral over $(\kappa_2, \kappa_\perp)$ of the axisymmetric spectrum. Defining the diagonal one-dimensional spectra along the line,

$$E_n^{\parallel}(\kappa_2) = \int_0^\infty \Phi_{nn}(\kappa_2, \kappa_\perp) 2\pi \kappa_\perp d\kappa_\perp \quad (\text{no sum}), \quad (5.5)$$

the rapid pressure–strain term reduces to a two-dimensional integral over the half-plane $(\kappa_2, \kappa_\perp)$ of the axisymmetric spectrum. The axisymmetric, solenoidal spectrum is fixed by two scalar functions of $(\kappa_2, \kappa_\perp)$ (Batchelor 1946; Chandrasekhar 1950); we denote them $a(\kappa_2, \kappa_\perp)$ and $c(\kappa_2, \kappa_\perp)$, where a is the isotropic-like scalar (the sole survivor when the turbulence is isotropic) and c measures the axisymmetric departure from isotropy. Carrying out the azimuthal average (5.4) and specialising to the plane strain $A = \text{diag}(\frac{1}{2}, -\frac{1}{2}, 0)$, the diagonal rapid pressure–strain components are

$$\Pi_{nn}^{(r)} = \int_0^\infty \int_0^\infty g_n(x) \overset{[x, a(\dots), c(\dots)]}{\left[a(\kappa_2, \kappa_\perp), c(\kappa_2, \kappa_\perp) \right]} 2\pi \kappa_\perp d\kappa_\perp d\kappa_2 \quad (\text{no sum on } n), \quad (5.6)$$

with $x \equiv \kappa_2^2 / \kappa^2$, $\kappa^2 = \kappa_2^2 + \kappa_\perp^2$, and the kernels, linear in the two spectral scalars,

$$g_1(x) = a\left(\frac{1}{8} + \frac{3}{4}x - \frac{7}{8}x^2\right) + c \frac{7}{8}x(1-x)^2, \quad (5.7)$$

$$g_2(x) = -\frac{3}{2}x \left[a(1-x) + c(1-x)^2 \right], \quad (5.8)$$

$$g_3(x) = a\left(-\frac{1}{8} + \frac{3}{4}x - \frac{5}{8}x^2\right) + c \frac{5}{8}x(1-x)^2. \quad (5.9)$$

Equations (5.6)–(5.9) constitute the axisymmetric collapse: *under A1, the rapid pressure–strain term is a functional of the axisymmetric spectral scalars a, c on the resolved half-plane and the mean strain*, with no reference to the unresolved azimuthal structure beyond what A1 fixes. The kernels satisfy two exact constraints that verify the reduction. First, energy conservation: pressure–strain redistributes without changing the trace, and indeed $g_1(x) + g_2(x) + g_3(x) = 0$ identically for all x and for both scalars, so $\Pi_{11}^{(r)} + \Pi_{22}^{(r)} + \Pi_{33}^{(r)} = 0$. Second, the isotropic limit: setting $c = 0$ with a a function of κ alone, the integral of g_3 over the half-plane vanishes, so $\Pi_{33}^{(r)} = 0$ —as it must, since the plane strain has no component along x_3 and the isotropic rapid term is proportional to S_{ij} (Crow 1968). Both constraints hold identically in the derived kernels, confirming the tensor reduction.

The status of (5.6)–(5.9) must be stated precisely, because it is here that the closure question is decided. Under A1 the rapid term is a linear functional of the two axisymmetric scalars $a(\kappa_2, \kappa_\perp)$ and $c(\kappa_2, \kappa_\perp)$ on the resolved half-plane—a *two-dimensional* representation, not yet a single-line closure, since the line determines only the $\kappa_\perp$ -integrals (5.5) of fixed combinations of a and c and not the functions themselves. Equations (5.6)–(5.9) are therefore the *collapsed form* on which any single-line statement must be built; the single-line content itself—what the line data pin down, and how sharply—is established next.

739

e	$Sk_t/\varepsilon = 0.8$				$Sk_t/\varepsilon = 16$			
	b_{22}	γ_{eff}	$m=2$	$\frac{\phi_{11}}{\phi_{33}} - 1$	b_{22}	γ_{eff}	$m=2$	$\frac{\phi_{11}}{\phi_{33}} - 1$
0	0.003	0.10	0.133	+0.06	0.003	0.10	0.133	+0.06
0.5	0.064	0.79	0.166	-0.20	0.070	1.28	0.247	-0.21
1	0.125	1.44	0.189	-0.32	0.124	4.22	0.396	-0.46

Table 2. A1 measured under plane strain (128³ DNS, four realisations): upwash anisotropy, effective polarisation, azimuthal $m = 2$ residue of Φ_{22} (isotropic floor 0.133), and the on-line falsifier T1.

5.4. What line data determine: sharp bounds on the rapid term

Under A1 the solenoidal spectrum tensor takes the two-scalar form $\Phi_{mn} = a P_{mn} + c \xi_m \xi_n$, with P_{mn} the solenoidal projector and ξ_m the projected axis direction; realisability is the pair of pointwise conditions $a \geq 0$ and $a + (1 - x) c \geq 0$ with $x = \kappa_2^2 / \kappa^2$. The line carries $\phi_{11}(\kappa_2) = \phi_{33}(\kappa_2)$ and $\phi_{22}(\kappa_2)$: per resolved wavenumber, two weighted $\kappa_\perp$ -integrals of the two unknown functions. The rapid term (5.6) is linear in (a, c) , so its extreme values over everything consistent with the line data, solenoidality and realisability are obtained from a family of linear programs, one per κ_2 , yielding sharp bounds $[\Pi_{nn}^-, \Pi_{nn}^+]$ and a kinematic indeterminacy $\mathcal{I}_n = (\Pi_{nn}^+ - \Pi_{nn}^-) / 2k_t S$ that quantifies exactly how much the null space costs. The forward map from (a, c) to the line spectra and the derivation of the bound kernels are given in Appendix A.

Calibrated on an isotropic von Kármán spectrum (with a verification suite pinning tracelessness, the isotropic angular averages $+\frac{1}{5} : -\frac{1}{5} : 0$, the longitudinal–transverse relation, and the exact $\frac{4}{5} k_t S_{ij}$), the indeterminacy is

constraint level	$\mathcal{I}_{11}$	$\mathcal{I}_{22}$	$\mathcal{I}_{33}$
line data + solenoidality + realisability	0.77	1.28	0.86
+ log-Lipschitz slope cap ($\lambda = 4$)	< 5% narrower		
+ polarisation cap $ c \leq \gamma a$, $\gamma = 0$	0.34	0.58	0.24

Two features of this table matter. Smoothness assumptions buy almost nothing: the null space is not a roughness artefact. The polarisation cap is the lever—but § 5.5 shows that under strain the effective polarisation reaches order one, so the honest bound in the strained regime is the uncapped one.

The representation also supplies two *free falsifiers* that the line itself reports: under A1, $\phi_{11}(\kappa_2) = \phi_{33}(\kappa_2)$ at every wavenumber (T1) and the velocity cross-spectra vanish (T2). Any measured transverse splitting is a wavenumber-resolved violation of A1; rather than assuming the hypothesis, the line monitors it.

5.5. The axisymmetry hypothesis, measured

A1 is exact for the isotropic inflow and is degraded by the plane strain itself. Rather than estimating this error perturbatively, we measure it in the strained-box DNS of § 4.6, where the full spectrum is available: the exact azimuthal decomposition of Φ_{22} yields the true scalars (a, c) , the discarded $m = 2$ azimuthal residue (the direct A1-violation measure, with isotropic sampling floor 0.133 in this box), and the effective polarisation $\gamma_{\text{eff}} = |c|/a$.

Three conclusions follow (table 2). First, the A1 error is *not* perturbatively small in the accumulated anisotropy: the residue grows from its floor to 0.19 at $Sk_t/\varepsilon = 0.8$ and to 0.40 at 16 by $e = 1$ —larger at rapid strain, because at moderate strain rate the nonlinear dynamics restore axisymmetry as fast as the strain destroys it. An $O(b^2)$ error estimate, which the symmetric structure of the discarded harmonic suggests, is superseded by this

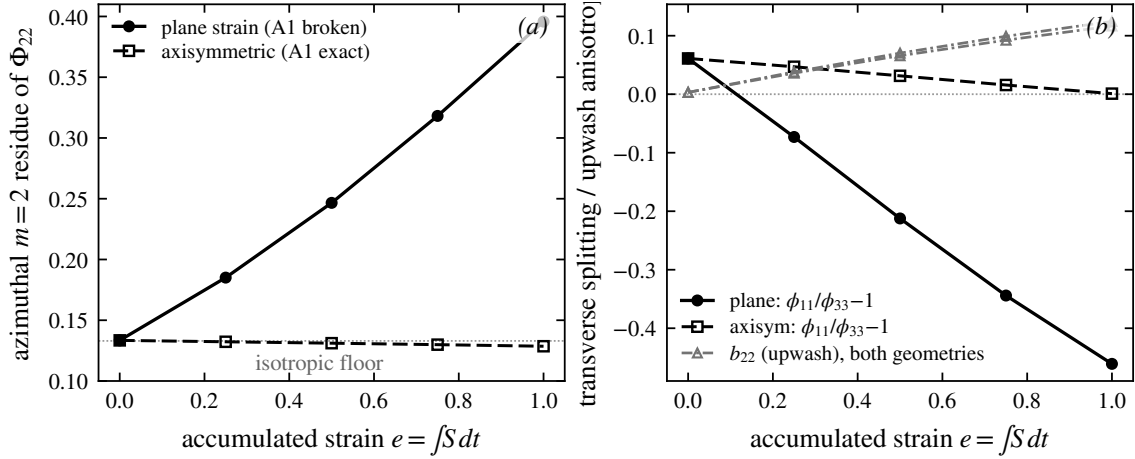


Figure 14. The A1 control experiment at $Sk_t/\varepsilon = 16$ (128³, four realisations): plane strain (A1 broken; solid, filled circles) against axisymmetric contraction at matched A_{22} (A1 exact by symmetry; dashed, open squares). (a) Azimuthal $m = 2$ residue of Φ_{22} : it grows under plane strain but stays at the isotropic floor (grey dotted) under axisymmetric strain. (b) Transverse splitting $\phi_{11}/\phi_{33} - 1$ per geometry, and the upwash anisotropy b_{22} (grey dash-dot, indistinguishable between geometries)—the azimuthal structure A1 assumes away is what differs, while the acoustically relevant upwash amplification is robust.

measurement. Second, the polarisation cap that narrows the isotropic bounds is illegitimate under strain: γ_{eff} passes unity by $e \approx 0.5$. Third, correspondingly, the uncapped bound brackets the exact rapid term (evaluated by direct integration in the DNS) up to $e \approx 0.25$ –0.5 and fails beyond, and the failure is A1’s, not the estimator’s.

The decisive evidence for that attribution is a control experiment: the same box under *axisymmetric contraction* with matched A_{22} , for which A1 is exact by symmetry. There the $m = 2$ residue stays at the sampling floor and the transverse splitting at zero at every strain up to $e = 1$ even at $Sk_t/\varepsilon = 16$, while under plane strain both grow (figure 14). The single-line closure is therefore *exact for axisymmetric distortion*; its plane-strain error is a strain-geometry property, now quantified, rather than a defect of the line representation. For the applications this maps cleanly: a rounded, three-dimensional stagnation region imposes an axisymmetric strain (closure exact); a two-dimensional leading edge imposes plane strain (closure carries the measured correction). The acoustically relevant amplification $b_{22}(e)$, finally, is indistinguishable between the two geometries (0.124 against 0.117 at $e = 1$, $Sk_t/\varepsilon = 16$): the quantity the leading-edge response consumes is robust to precisely the geometry that A1 is sensitive to.

The only previous mention of LRR-QI is in Fig. 7, with no explanation of how it differs from LRR.

5.6. Component ordering under plane strain

The exact solution of the rapid problem under plane strain has the neutral spanwise component eventually overtaking the upwash; algebraic closures predict the opposite. The exact rapid term evaluated in the DNS settles the matter: at $e = 1$, $\Pi_{33}^{(r)}/k_t S = 0.14$ at $Sk_t/\varepsilon = 0.8$ and 0.27 at 16, exceeding $\Pi_{11}^{(r)} = 0.23$ in the rapid case—the overtaking is real and sets in at rapid strain. LRR-QI, whose spanwise component is carried by its b -dependent term, matches at $Sk_t/\varepsilon = 0.8$ ($\Pi_{33}^{(r)} = 0.143 k_t S$) but delivers half the exact value at 16; IP gives zero identically. The deficiency thus lives in the moment closure’s b -term, not in the line model: the ODT moment-level b_{ij} follows the closure to $\sim 2\%$, and its b_{33} stays within 0.006 of zero in every run. In the spectral representation the ordering is carried by the c -scalar through the kernels (5.7)–(5.9); the moment reduction discards

it. This answers, by measurement, where the known component-ordering error enters and what it would take to remove it.

5.7. Reduction to the moment closure

The closed form (5.6)–(5.9) operates on the resolved spectrum, whereas the distortion results of § 4 were obtained with the algebraic moment closure of § 2.4 (Launder *et al.* 1975). For the two to be consistent, the spectral kernel must reduce to that moment closure in the limit where the moment closure is exact—isotropic turbulence. We show that it does, which both justifies the closure used in the simulations and fixes the prefactor in an explicit convention. Where is this done?

Figure 15 plots the kernels g_1, g_2, g_3 of (5.7)–(5.9) against the angular variable $x = \kappa_2^2/\kappa^2$. Two features are exact and hold for both spectral scalars. The kernels sum to zero at every x ,

$$g_1(x) + g_2(x) + g_3(x) = 0 \quad \forall x, \quad (5.10)$$

so the rapid term conserves turbulent kinetic energy pointwise in wavenumber, not merely in the integral; and each kernel vanishes at $x = 1$, where the wavevector aligns with the line and the projection $\kappa_a \kappa_b / \kappa^2$ degenerates. The signs are physically ordered: $g_1 > 0$ (energy fed to the streamwise component E_1), $g_2 < 0$ (energy removed from the line/upwash component E_2) over the bulk of the angular range, with g_3 changing sign, consistent with the plane strain stretching x_1 and compressing x_2 .

For *isotropic* turbulence the anisotropy scalar vanishes ($c = 0$) and the isotropic-like scalar a depends only on κ . Averaging the kernels over solid angle—equivalently, integrating $g_n(x)$ against the isotropic distribution of x on the sphere—gives

$$\langle g_1 \rangle : \langle g_2 \rangle : \langle g_3 \rangle = +\frac{1}{5} : -\frac{1}{5} : 0, \quad (5.11)$$

in the ratio $+1 : -1 : 0$. This is exactly the structure of the plane strain $S_{ij} = \text{diag}(\frac{1}{2}, -\frac{1}{2}, 0)$: the rapid term is proportional to S_{ij} , with $\Pi_{33}^{(r)} = 0$ because the strain has no component along x_3 . The spectral kernel therefore collapses, under isotropy, to the isotropic-production form of the moment closure,

$$\Pi_{ij}^{(r)} = \frac{4}{5} k_t S_{ij} \quad (\text{isotropic limit}), \quad (5.12)$$

the result of Crow (1968) that underlies the isotropisation-of-production (IP) rapid term in the closure of Launder *et al.* (1975). We note a convention: with the symmetrised definition of the pressure–strain correlation adopted here and the factor of two carried in the Poisson source (2.6), the kernels (5.7)–(5.9) yield the coefficient $\frac{2}{5}k_t$ for the 11-component directly (since $S_{11} = \frac{1}{2}$), equivalent to the standard $\frac{4}{5}k_t S_{ij}$ of (5.12); the ratio (5.11) and the tracelessness (5.10) are independent of this convention.

The consequence is that the algebraic closure used to obtain the distortion of § 4 is not an independent assumption but the isotropic-limit reduction of the line-consistent spectral term derived here. The spectral form (5.6) carries, in addition, the angular (x -dependent) and anisotropy (c -scalar) information that the moment closure discards—and both discards are now measured: the wavenumber dependence of the linear response in the three-way comparison of § 4.6.1, and the component ordering in § 5.6. The moment closure and the collapsed form thus agree to leading order, and the terms in which they depart are exactly the measured deficiencies of § 4.

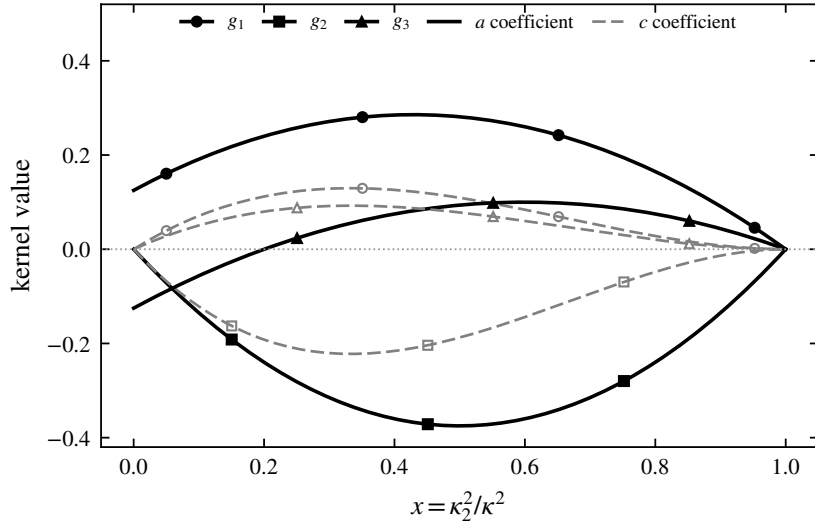


Figure 15. Rapid pressure–strain closure kernels g_1 (circles), g_2 (squares), g_3 (triangles) (eqs. (5.7)–(5.9)) against the angular variable $x = \kappa_2^2/\kappa^2$. Solid curves: coefficient of the isotropic-like spectral scalar a ; faint grey dashed curves: coefficient of the anisotropy scalar c . The kernels sum to zero at every x (energy conservation, eq. (5.10)) and vanish at $x = 1$. Averaged over an isotropic spectrum they give the ratio $+\frac{1}{5} : -\frac{1}{5} : 0$ (eq. (5.11)), recovering the isotropisation-of-production form $\Pi_{ij}^{(r)} = \frac{4}{5} k_t S_{ij}$ (Crow 1968) that underlies the moment closure used in § 4.

5.8. Consistency with the eddy redistribution: avoiding double counting

A closed rapid term is necessary but not sufficient: it must not duplicate redistribution already effected by the eddy events. The pressure–strain correlation decomposes canonically into rapid and slow parts, $\Pi_{ij} = \Pi_{ij}^{(r)} + \Pi_{ij}^{(s)}$, corresponding to the two source terms in (2.6). The slow part $\Pi_{ij}^{(s)}$ is the return-to-isotropy mechanism, quadratic in the fluctuations and independent of the mean gradient; the eddy events in ODT, which act on the fluctuating field without reference to A_{kl} , are the model’s representation of exactly this slow redistribution. The rapid part $\Pi_{ij}^{(r)}$, proportional to A_{kl} and linear in the fluctuations, is by construction absent from the eddy mechanism, which carries no dependence on the mean strain. The two therefore occupy disjoint terms of the canonical decomposition, and adding (5.6) as an explicit mean-strain-driven source alongside the eddies does not double-count, provided the eddy events are calibrated to reproduce the slow return-to-isotropy in the absence of mean strain—a condition that the no-strain baseline of § 4.3 verifies directly, since there the centroid relaxes under the eddies alone with no spurious rapid contribution. This argument establishes consistency at the level of the decomposition; the no-strain baseline of § 4.3 provides the direct numerical confirmation, the centroid there relaxing under the eddy mechanism alone.

The disjointness argument is now also a measured statement, from both sides. With the strain switched off the eddy events generate no spurious strain-like signal (the nominal-turbulence null of Appendix B). Conversely, in the rapid limit ($Sk_t/\varepsilon = 16$, under one eddy per line in $e \leq 1$) every kernel-energy allocation rule tested—the isotropic kernel, a strain-biased generalisation, and discrete eddy types—reproduces the exact rapid-distortion evolution of $b_{22}(e)$ to ± 0.003 : the rapid term cannot be reached from inside the eddy events, and the continuous operator carries it alone. Rapid and slow redistribution occupy disjoint terms by construction, and now by measurement.

5.9. Summary: what the single-line treatment establishes

1. The rapid term is exactly $A_{kl}M_{ijkl}$; its single-line representation is obstructed by an azimuthal null space that is stated exactly and survives the axisymmetric collapse at the level of the two spectral scalars.
2. Under A1 the term collapses to the kernel form (5.6)–(5.9); A1 is exact for axisymmetric distortion and is *measured* under plane strain, where its violation grows with accumulated strain and with strain rapidity (table 2).
3. Independently of A1's validity, ^{DNS} line data bound the rapid term sharply; the bound is honest—and verified to bracket the exact term—up to $e \approx 0.25$ – 0.5 under plane strain, and at all computed strains under axisymmetric distortion.
4. The moment closure used in the simulations is the isotropic reduction of the collapsed form (verified, including the exact $\frac{4}{5}k_t S_{ij}$); what it discards is the wavenumber dependence of the linear response and the component ordering, both measured in § 4 and § 5.6.
5. Rapid and slow redistribution are disjoint: verified by the strain-off null and by the rapid-limit invariance of every ^{tested} kernel-energy allocation rule.
6. The acoustically relevant upwash amplification $b_{22}(e)$ is linear to $e = 1$ and robust to the strain geometry that controls A1's validity.

6. Measured limits, and scale-local relaxation

The envelope of § 4.6.2 localises the model's spectral deficiency in its wavenumber-uniform rapid kinematics: the strain-induced anisotropy is allocated uniformly across scales ^{where} while linear theory and DNS concentrate it at the energy-containing range and remove it from the fine scales. This section asks the constructive question: which mechanism available to the model can supply the missing scale dependence? We answer it experimentally, by a scale-resolved diagnostic and a controlled family of interventions in the eddy mechanism, each tested at 1024 realisations with seeds paired to a common baseline, and each either falsified or confirmed by the same measurement. The outcome is a single measured statement: what is missing is *scale-local relaxation*, and supplying it—at the scales of the imbalance—is both necessary and, at moderate strain, sufficient to move the fine-scale anisotropy.

6.1. A transmission diagnostic

The configuration is a scale-diagnostic variant of the strained line: unit strain rate (so $e = t$), an initial spectrum peaked at a wavenumber κ_p resolved by more than a decade of smaller scales, and two measurement bands, $\kappa_2 \in [0.6, 2] \kappa_p$ (the energy-containing band) and $[6, 16] \kappa_p$ (roughly two triplet-map generations below it, since each map compresses by a factor of three). On each band ^b the component imbalance is $\mathcal{A}_b = E_2^\parallel(b) / \frac{1}{2} [E_1^\parallel(b) + E_3^\parallel(b)]$, the band-integrated upwash energy over the transverse mean; all statistics are medians over realisations with bootstrap confidence intervals, for the intermittency reason given in § 4.6. The model's eddy ^{boosts} and kernel mechanisms are component-symmetric, so $\mathcal{A} = 1$ is their fixed point; the strain ^{drives} $\mathcal{A}$ up at the scales where it deposits anisotropy, and the ratio $\mathcal{A}_{hi}/\mathcal{A}_{lo}$ measures how much of the large-scale imbalance survives transmission down the cascade. ^{Until the last sentence of this section it's unclear whether these refer to length scale or wavenumber.}

The baseline (supplementary material, figure S3, black; also the black curve of figure 16) exhibits the deficiency of § 4.6.1 in transmission form: $\mathcal{A}_{hi}/\mathcal{A}_{lo}$ stays near unity out to

905 $e = 3$ (1.02, 0.95, 0.89 at $e = 1, 2, 3$)—the anisotropy injected at the energy-containing
 906 scales crosses two map generations essentially undiminished—and drops only at the deepest
 907 strain. The mechanism is direct: the high band is populated by the maps of *large* eddies
 908 (one generation takes κ_p to $3\kappa_p$, a second to $9\kappa_p$, inside the band), so fine-scale anisotropy
 909 is delivered by large-eddy events, not by a chain of small ones.

910 This lacks context, indicating that some framing of the
 investigation is needed at the beginning of this section.

911 6.2. Interventions: what fails and what works

912 The complete intervention family—each case’s knob, measured fine-scale effect and
 913 reading, with the four-panel diagnostic figure for the eddy-locked variants—is tabulated in
 914 the supplementary material (§ S2, table S1 and figure S3); the ledger it establishes is set
 915 out here, with the two decisive figures (the relaxation clock and the five-way benchmark)
 916 retained below. Two families fail for instructive reasons. Acceptance gating—rejecting
 917 accepted eddies whose post-event state remains anisotropic, at thresholds spanning 45
 918 to 98 per cent rejection—is scale-blind: at mild thresholds the surviving eddy population
 919 reproduces the baseline anisotropy budget in every band (the gate self-selects eddies whose
 920 kernels were already going to do the relaxing, and the unrelaxed field keeps generating
 921 candidates), while at the harshest threshold the model merely drifts towards its eddy-free
 922 limit. Restricting the gate to sub-band-scale eddies sharpens the negative result into the
 923 mechanism statement above: removing up to 97 per cent of small-eddy events leaves both
 924 the fine-scale anisotropy *and* the fine-scale energy unchanged, so no acceptance-gating
 925 scheme can control transmission at **this Reynolds number**. Reweighting the triplet-map
 926 images (a middle image of relative size $\frac{2}{3}$, which halves the compression of the map’s
 927 dominant central channel) moves one-point and energy-containing statistics— $\mathcal{A}_{10}$ falls
 928 significantly and the upwash energy fraction with it—but not the fine scales: with the
 929 image fractions summing to one, slowing the middle channel necessarily speeds the outer
 930 ones, and one outer image maps the energy peak directly into the measurement band.

931 What works is relaxation applied *at the scale of the imbalance*. After each accepted
 932 eddy’s map and kernels, the same energy-equalising kernel procedure (§ 2) is applied to
 933 each of the three map images—the sub-interval’s own triplet-map displacement shape as
 934 the kernel, coefficients from the sub-interval’s own integrals, so that each sub-application
 935 conserves the momentum components and the total kinetic energy exactly, with no map
 936 applied and no random numbers drawn—and then recursively to 9 and 27 sub-regions.
 937 One **level** ($\ell/3$ for an eddy of size ℓ) is **null**; for the large eddies that feed the band, $\ell/3$ still
 938 sits above it. Two **levels** produce the first statistically significant fine-scale isotropisation
 939 of the family ($\mathcal{A}_{hi}$ from 1.43 to 1.27 at $e = 1$, **both bands’ paired contrasts individually**
 940 significant), and three **levels** extend the effect to deep strain: the **levels that work, $\ell/9$**
 941 **and $\ell/27$** , are exactly the **first** that reach the measured band. The **bookkeeping is clean**
 942 throughout—band energies move by a few per cent, the upwash energy fraction and
 943 the intermittency tails are unchanged—and the **pre-strain fine-scale floor** moves towards
 944 component equality immediately ($\mathcal{A}_{hi}$ from 0.83 to 0.95 at $e = 0$).

945 **Four** further variants **separate the candidate** explanations for the one remaining limitation,
 946 the fading of the gains beyond $e \approx 3$. Tying the recursion depth to the eddy size (subdividing
 947 while the sub-interval remains above a near-dissipative floor, so that large eddies do
 948 proportionally more sub-scale work) gives the gains of depth with no reversal at the
 949 deepest strain: both bands significant at $e = 1$ and the effect retaining its sign through $e = 3$.
 950 Iterating the two-level pass three times within each event answers the amplitude hypothesis
 951 in the negative: it is a stronger early isotropiser but overshoots at deep strain. Delocalising
 952 the relaxation—after each eddy, applying the kernel isotropisation in N randomly placed
 953 intervals of the same size, overlaps allowed—tests whether the *location* of the events is the

inconsequential

limiter: $N = 1$ is **null**, and $N = 3$ is significant in both bands at $e = 1$ but acts mainly at the energy-containing scales, since the sampled intervals inherit the eddy-size distribution and full-interval isotropisation acts at that scale. Stacking the two ideas—the adaptive hierarchy inside each sampled interval—is the strongest early isotropiser of the family ($\mathcal{A}_{hi}$ from 1.43 to 1.26 at $e = 1$) and delivers the only significant deep-strain effect, in the energy-containing band at $e = 3.9$; yet at the fine scales and deep strain it overshoots the depth-only variant significantly. **The ledger is then complete** for every scheme locked to the eddy clock: depth sets *where* the relaxation acts and is necessary; dosage and delocalisation raise the early gains in proportion to the total relaxation delivered; and no eddy-locked combination—more depth, more dosage, more events, other locations—prevents the fine-scale fade beyond $e \approx 3$. The remaining suspect is the clock itself: relaxation acts at the eddy rate while the strain resupplies the imbalance continuously. **Whether that is fundamental or merely structural is decided by** the final intervention.

This motivates

The candidate distribution is not the size distribution of implemented eddies, nor does it have its own theoretical justification. Its functional form should be shown with the justification that it is convenient to sample and has the gross qualitative features of the physical eddy-size distribution. 'uncapped' and 'size-capped' are meaningful only with reference to the parameters of this distribution.

6.3. Decoupling the relaxation clock

feature

formulations do not modify

The one **axis** the eddy-locked **family cannot test** is *when* relaxation acts. We therefore give it its own clock: an independent Poisson stream of relaxation-only events at a prescribed rate, each event an interval drawn from **the model's eddy-size distribution**, placed uniformly on the domain, and relaxed by the same conservative kernel pass with the adaptive-depth hierarchy inside it. **Because each event acts on an interval**, momentum and kinetic energy are conserved exactly per event; because the stream **has its own rate**, its **pacing** is decoupled from the eddy statistics **entirely**. The **rate** is matched to the **baseline's** mean eddy rate, so the comparison against the eddy-locked variants is dose-matched by construction.

The result (figure 16) decides the question. With the clock, the fine-scale imbalance is reduced significantly at *every* strain and the effect grows rather than fades—the paired contrasts reach -0.30 (low band) and remain significant in the high band at $e = 3.9$ —because a constant-rate stream keeps pace precisely where the eddy rate could not: event-locked timing, not any per-event property, was the deep-strain limiter. But the **uncapped** stream is scale-blind and overcorrects: it also suppresses the one-point anisotropy that the strain should produce, pulling $u_2^2/2k_t$ from 0.58 to 0.41 at $e = 3.9$ and thereby breaking the moment-level behaviour validated in § 3.4. Restricting the stream to sub-band interval sizes ($\ell \leq \ell^*$, the scale separating the two measurement bands) resolves the tension: the size-capped clock—the “clock-relaxed ODT” of the figures—holds the fine scales at every strain with no fade— $\mathcal{A}_{hi}$ at 1.30–1.44 against the baseline's 1.63–1.94, **contrasts** significant throughout and strongest at $e = 3$ —while the **one-point trajectory** is essentially untouched in the moderate-strain regime ($u_2^2/2k_t$ within 0.005 of the baseline at $e = 1$) and only mildly reduced at the deepest strain. The **rate is a clean dial**; **four times** the rate **holds the fine scales harder** at a proportionate **one-point cost**.

extends the upward trend in Fig. 16(a)

Cost in what sense? No physical reference point is discussed.

in Fig. 16(b)

mean rate of relaxation events directly controls the strain dependence: quadrupling of

6.4. The measured missing ingredient

As on p. 31, I don't understand this.

The arc closes the loop opened by the three-way comparison. The deficiency measured there—anisotropy allocated uniformly over wavenumber—is confirmed here **in trans-mission form**, is unfixable by gating or map reweighting, and is cured by exactly two ingredients, each isolated by a falsified alternative: relaxation applied *at the scale of the imbalance* (depth, not dosage or location) and relaxation applied *on its own clock* (the size-capped **stream**, which alone survives deep strain). One limit remains, and it is structural rather than curable by any amount of relaxation: an isotropising mechanism is bounded at component equality, whereas exact linear theory and the DNS carry the fine-scale upwash

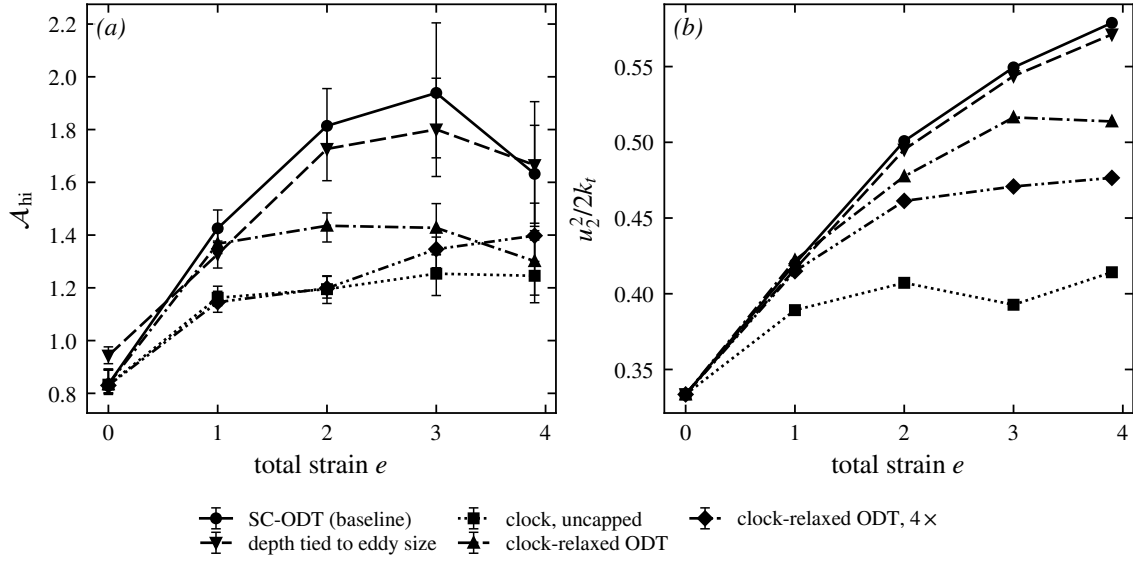


Figure 16. The independent relaxation clock against the eddy-locked family. (a) Fine-scale-band imbalance $\mathcal{A}_{hi}$ against total strain: the eddy-locked variants fade beyond $e \approx 3$; the clock holds at every strain. (b) The price: upwash energy fraction—the uncapped clock suppresses the physical one-point anisotropy, the size cap restores it. Medians over 1024 paired-seed realisations; the clock rate is matched to the baseline eddy rate.

Is the one-point anisotropy distinctly more physical than anything else, especially considering that there is no comparison here to actual physical values? or, for the 4X case, multiplied by a factor of 4.

below the transverse mean. Crossing that boundary requires the rapid operator itself to discriminate by wavenumber, and the formulation already contains its natural form: the spectral kernel (5.6)–(5.9) is precisely a wavenumber- and angle-dependent rapid operator, whose discretisation in place of the wavenumber-uniform moment closure of § 2 is the identified next step.

The ^{comparison} head-to-head that summarises this part is figure 17: the strain-induced upwash anisotropy per wavenumber band—the observable on which § 4.6.1 established the benchmark—for exact linear theory, the DNS, SC-ODT, and the model under each of the two mechanism forms (adaptive depth; the size-capped clock), at three strain-rate parameters, from same-binary ensembles under the identical precursor protocol. Four statements can be read directly from the shared axes. First, at $Sk_t/\varepsilon = 16$ the DNS lies on the exact-RDT curve: the rapid limit is realised in the benchmark itself, and the deficiency of the baseline model—its uniform allocation of a correct integrated anisotropy—is starkest precisely where the eddies matter least. Second, at moderate rapidity the mechanism tilts the model’s allocation towards the DNS—the anisotropy now decreases with wavenumber, the fine-band excess roughly halved (at $Sk_t/\varepsilon = 0.8$: from +0.076 against the DNS value −0.046 to +0.024)—and the improvement grows as the rapidity falls, the trend the application regime sits on. Third, within $e \leq 1$ the two mechanism forms are equivalent on these axes; the clock’s advantage is the deep-strain hold of figure 16, which this protocol—bounded at $e = 1$ by the DNS reference—cannot display. Fourth, at $Sk_t/\varepsilon = 16$ neither form moves the allocation: rapid strain limits the physical time available, and no relaxation scheme manufactures more of it. The remaining, sign-level gap belongs to the wavenumber-dependent operator identified above.

With so many mechanisms at play, this should be referenced more explicitly.

7. Consequence for leading-edge noise

The application that motivates the model can now be served with its output. In the leading-edge-noise theory of Amiet (1975) the far-field pressure spectral density of a large-aspect-

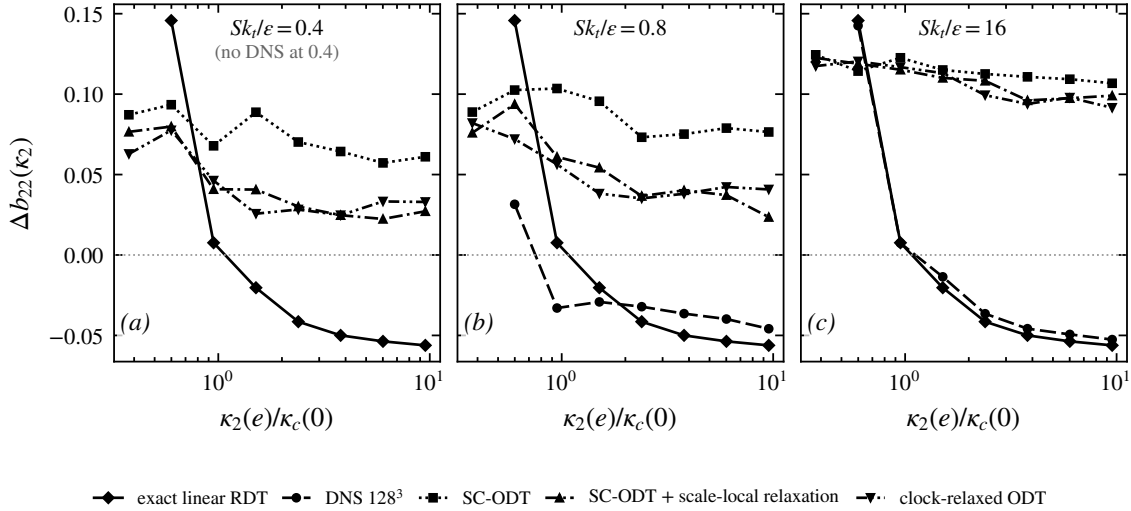


Figure 17. The five-way comparison on the benchmark observable of § 4.6.1: strain-induced upwash anisotropy per wavenumber band at $e = 1$, for exact linear RDT, DNS, SC-ODT, and SC-ODT with each mechanism form—scale-local relaxation at adaptive depth, and the clock-relaxed variant (size-capped relaxation clock)—at $Sk_t/\varepsilon = 0.4, 0.8$ and 16 (ODT: 1024 realisations per case, same binary and precursor protocol; DNS and companions as in § 4.6; no DNS exists at 0.4 , where exact RDT is the reference—at fixed total strain it is independent of the strain rate). At rapid strain the DNS realises the exact-RDT allocation; the mechanism moves the model towards the DNS increasingly as the rapidity falls.

ratio aerofoil factorises as a product of the flow and observer geometry, the aerofoil response $|\mathcal{L}|^2(K_x)$, the upwash wavenumber spectrum $\Phi_{ww}(K_x, 0)$ and the spanwise correlation length $\ell_y(K_x) = \pi \Phi_{ww}(K_x, 0) / \int \Phi_{ww}(K_x, K_y) dK_y$, with K_x the streamwise and K_y the spanwise wavenumber. For a *change of inflow spectrum* at fixed aerofoil, flow and observer, everything except the turbulence kernel

$$\mathcal{K}(K_x) = \pi \frac{\Phi_{ww}(K_x, 0)^2}{\int \Phi_{ww}(K_x, K_y) dK_y} \quad (7.1)$$

cancels exactly, so the change of radiated level is $\Delta \text{SPL}(K_x) = 10 \log_{10}[\mathcal{K}^{\text{distorted}}/\mathcal{K}^{\text{frozen}}]$, independent of the aerofoil geometry and response. This is the sharpest question the distortion model can be asked without importing an acoustic solver: by how much does the prediction change when the inflow spectrum is the *computed, distorted* one instead of the frozen upstream spectrum that standard practice prescribes?

The distorted input is taken from the model's own strained ensembles in the representation already fitted in § 4.6.2: the exact-RDT-distorted von Kármán family, two parameters (A_0, κ_e) per strain, fitted to the median line spectra of the 1024-realisation ensembles at $Sk_t/\varepsilon \approx 0.4$ and ≈ 8 . The line data pin the resolved-wavenumber structure; the transverse structure that (7.1) requires is supplied by the representation—this is § 5's statement about line knowledge, operating in the application—and the representation's rms residual, 13–24% over the strain range (the distance envelope of figure 13), is the corresponding uncertainty carried by the result. The two-dimensional spectrum follows by direct integration of the distorted components, $\Phi_{ww}(K_x, K_y; e) = A_0 \int \Phi_{22}^{\text{RDT}}(K_x, \kappa_2, K_y; e) d\kappa_2$, with the machinery verified against the wavevector-ensemble integrator of § 3.2 (upwash-variance growth reproduced to under 1%). The frozen baseline is the same inflow's relaxed $e = 0$ state carried undistorted. The level and shape contributions are separated: the kernel scales linearly with the upwash variance, so dividing it out isolates the spectral reallocation, and the variance ratio itself is taken from the measured line spectra rather than from the fit.

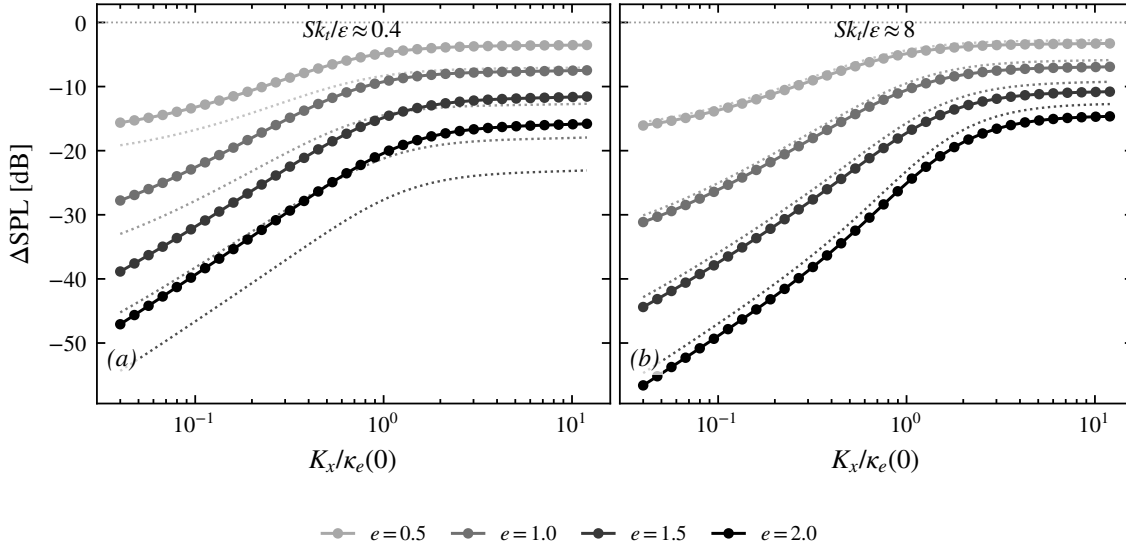


Figure 18. Change of the Amiet leading-edge-noise kernel when the frozen inflow spectrum is replaced by the computed distorted one, against reduced streamwise wavenumber, at total strains $e = 0.5$ – 2 (shade light to dark with increasing strain): (a) $Sk_t/\varepsilon \approx 0.4$, (b) ≈ 8 . Solid: spectral reallocation at matched upwash variance (shape only); dotted: total, including the measured variance change (–3.5 to –7.3 dB at the slow rate, where decay dominates the transit; +0.5 to +1.9 dB at the rapid rate, where the upwash amplification survives). Geometry, aerofoil response and observer position cancel identically in this ratio.

Does transit mean transient evolution?

Figure 18 shows the result, and its sign and structure are the finding. At matched upwash variance the distorted spectrum is a substantially *weaker* leading-edge source than the frozen one at every resolved reduced frequency: ≈ -5 dB at $K_x \approx \kappa_e(0)$ for $e = 0.5$, deepening to -20 dB and beyond by $e = 2$, and increasingly severe towards low K_x . The mechanism is the orientation physics of § 5 acting on the noise kernel: (7.1) weights the spanwise-coherent upwash content $\Phi_{ww}(K_x, 0)$, which lives on wavevectors aligned with the line—exactly the orientations whose upwash is smallest by solenoidality and whose pre-images retreat into the vanishing low-wavenumber tail of the initial spectrum as the distortion proceeds. The distortion, in short, drains the very content the aerofoil responds to. The total change adds the variance term: at the rapid rate the surviving upwash amplification offsets a small part of the reallocation; at the slow rate viscous decay over the longer transit deepens it.

undefined: cite a reference?

Figure 19 places the kernel result in the quantity practice actually reports: the absolute far-field spectrum of the reference configuration of § 4.4, assembled from the compact lift-dipole form of Amiet’s theory with the exact Sears response. The spectra enter in physical units through two matching conditions only—the integral scale fixes the length unit through the von Kármán relation $\kappa_e = 0.7468/\mathcal{L}$ and the upwash variance at $e = 0$ fixes the amplitude—so the frozen baseline is the standard Amiet–von Kármán prediction, not a recalibration of it: the $e = 0$ member of the representation is the isotropic von Kármán spectrum itself, and it reproduces the analytic one-dimensional form to within the quoted fit residual (ratio 0.79–1.14 across 100 Hz–1 kHz), with the Liepmann spectrum shown as an independent check of the unit bridge. Against this baseline the computed distortion lowers the predicted level by 5 dB at $e = 0.5$, 9 dB at $e = 1$ and 14 dB at $e = 2$ across the resolved band ($f \lesssim 1.8$ kHz at this operating point), the spanwise-coherence drain of figure 18 now carried through geometry, aerofoil response and radiation to the observer.

The chain is then confronted with measurement. Figure 20 shows four published leading-edge-noise experiments with fully documented inflow turbulence—the NACA0012 of

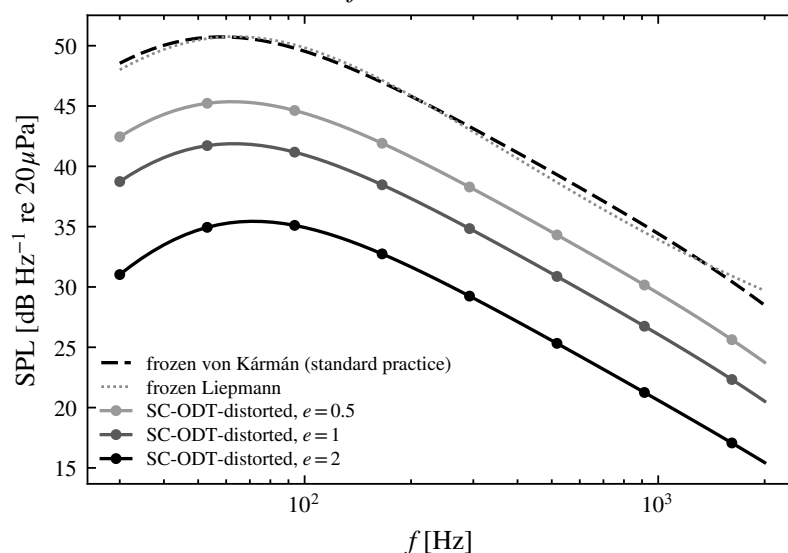
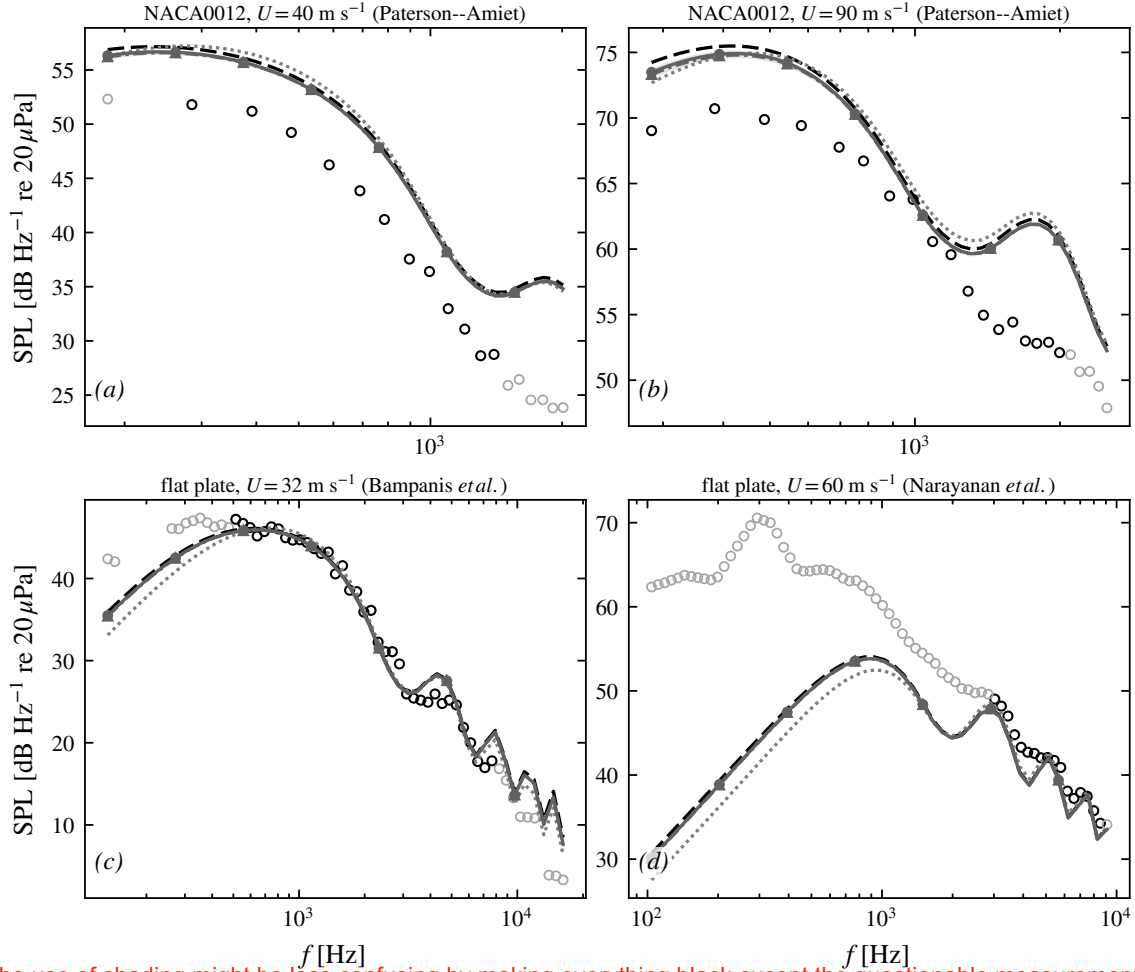


Figure 19. The assembled absolute prediction at the reference configuration of § 4.4: far-field sound pressure level from the compact lift-dipole form of Amiet (1975) with the exact Sears response, for a flat plate of chord 0.39 m ($Re_c \approx 5 \times 10^5$) and span 0.8 m at $U_\infty = 20 \text{ m s}^{-1}$, $u'/U_\infty = 6\%$, $\mathcal{L} = 55 \text{ mm}$, observer 1 m overhead. Dashed: the frozen von Kármán inflow of standard practice (the $e = 0$ member of the fitted representation, which is exactly the isotropic von Kármán spectrum); grey dotted: the frozen Liepmann spectrum at the same $(u', \mathcal{L})$, an independent analytic cross-check of the physical-unit bridge. Solid (grey light to black with increasing strain, circles): the same chain with the SC-ODT-distorted inflow at total strains $e = 0.5, 1$ and 2 ($Sk_t/\varepsilon \approx 0.4$), level and shape both included.

Why not also in Figs. 18 and 19?

Paterson & Amiet (1976) at $40\text{--}90 \text{ m s}^{-1}$ (higher speeds excluded only by their Mach number), and the flat plates of Bamptonis *et al.* (2022) and Narayanan *et al.* (2015)—each with four predictions computed through the identical response: the frozen von Kármán and Liepmann inflows of standard practice, and the model's computed spectra, SC-ODT and clock-relaxed ODT, evaluated at the effective free-distortion strain accumulated along the potential-flow stagnation streamline down to the eddy scale ($e_{\text{eff}} = 0.09\text{--}0.16$ for the NACA0012; below 0.05 for the thin plates). Three statements result. First, the assembled chain is quantitatively right where the experiment is cleanest: on the flat-plate cases every formulation sits within 1–2 dB rms of the measured spectra with the non-compactness interference lobes reproduced—the fidelity Bamptonis *et al.* (2022) report for their own implementation—so the comparison discriminates inflow physics, not implementation. Second, the computed distortion behaves causally: for the thin plates the effective strain is negligible and the model correctly predicts *no* correction to the frozen spectrum, while for the NACA0012 it lowers the prediction towards the data at every speed (bias $+5.7 \rightarrow +5.3 \text{ dB}$ at 40 m s^{-1} , $+1.8 \rightarrow +1.4 \text{ dB}$ at 60, $+5.1 \rightarrow +4.7 \text{ dB}$ at 90), by the modest amount its geometric effective strain implies; the shift between predictions is exact, since the digitisation uncertainty ($\pm 0.5 \text{ dB}$) is common to all four. Third, the SC-ODT and clock-relaxed ODT models are indistinguishable in the far field at these laboratory strains (differences below 0.05 dB), and this is stated as a limit of the test, not of the mechanism: laboratory rigs are designed to minimise inflow distortion, so no published far-field measurement samples the strains at which the relaxation mechanism acts. The residual NACA0012 overprediction is the finite-thickness effect Paterson & Amiet (1976) themselves identified (breakdown near $U/ft_A \approx 1$) and Gershfeld (2004) modelled, of a piece with the blocking mechanism excluded from the present scope; it is unchanged in rank by any choice of inflow spectrum.

There are multiple model variants, so this is ambiguous



The use of shading might be less confusing by making everything black except the questionable measurements. Reduced line thicknesses might also help.

○ measured — — Amiet + von Kármán Amiet + Liepmann —●— Amiet + SC-ODT —▲— Amiet + clock-relaxed ODT

Figure 20. Absolute far-field predictions against published measurements, all four inflow spectra through the identical non-compact response (Amiet 1975; Roger & Moreau 2010): frozen von Kármán (black dashed), frozen Liepmann (grey dotted), the computed SC-ODT spectrum (grey, circles) and the computed clock-relaxed ODT spectrum (grey, triangles), each entered at the case's effective free-distortion strain. (a,b) NACA0012 of Paterson & Amiet (1976) ($c = 0.23$ m, $u'_w/U \approx 4\text{--}5\%$, $\Lambda_f \approx 30$ mm, observer 2.25 m overhead; Gershfeld thickness admittance applied identically to all four chains); (c) ECL flat plate of Bampanis *et al.* (2022) ($c = 0.10$ m, 3 mm thick, $Tu = 4.5\%$, $\Lambda = 9$ mm, $\Theta = 90^\circ$, $R = 1.25$ m); (d) ISVR flat plate of Narayanan *et al.* (2015) ($c = 0.15$ m, $Tu = 2.5\%$, $\Lambda = 6$ mm, $R = 1.2$ m). Black measured symbols lie in each case's stated comparison band (grey outside: background flags, facility contamination, thickness-breakdown frequency); spectra digitised from the published figures with per-figure calibration checks (± 0.5 dB). The effective strain is a geometric estimate (potential-flow stagnation-line accumulation truncated at the eddy scale); the shaded band spans truncation at $\Lambda/4\text{--}\Lambda$.

Where?

Where the mechanism does reach the prediction is shown in figure 21, and the claim is worded exactly as the evidence permits. At total strains of 1–2 the frozen-spectrum error against the linear reference grows from +1.3 to +3.6 dB, while both computed inflows track the reference within 1–2.5 dB—a residual that contains the physical relaxation at finite rapidity as well as the model deficiency measured in § 4.6.2—so an evolved inflow beats a prescribed one at precisely the strains where the correction matters, whichever variant of the model evolves it. Between the two variants the radiated level does not discriminate: the clock acts below its size cap, where the noise kernel carries little weight,

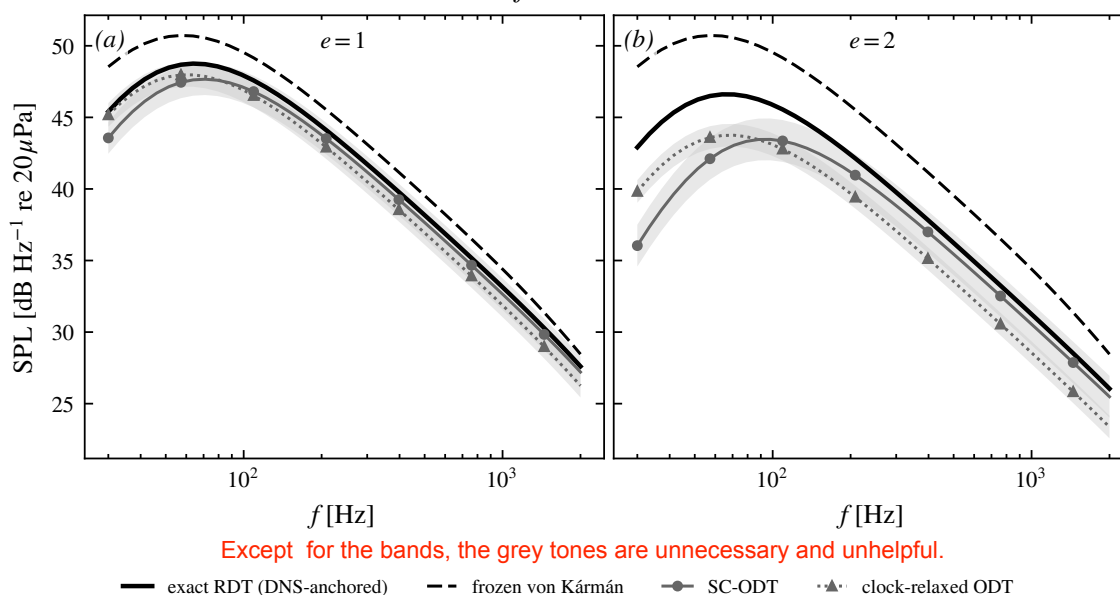


Figure 21. Computed against prescribed inflow at application strains, judged by the linear reference: absolute SPL at the reference configuration for the rapid family ($Sk_t/\varepsilon \approx 8$) at (a) $e = 1$ and (b) $e = 2$. Solid black: exact (inviscid) RDT of the model's relaxed initial state—the reference the strained-box DNS of § 4.6 validates at rapid rates—with level set by the kernel's exact upwash amplification (reproducing the moment-level value to 0.7%). Black dashed: the frozen von Kármán inflow of standard practice. Grey (circles: SC-ODT; triangles: clock-relaxed ODT): the two computed inflows (fitted representations, measured ensemble levels), each with a band giving its measured spectral rms distance from the RDT family mapped to decibels; the two share the band and are indistinguishable at this level.

bands overlap sufficiently so that the two inflows

and its gain appears instead in the uncertainty the input carries into the prediction—the clock-relaxed ensembles lie closer to the exact-RDT-distorted family by a factor of two to four in rms at both rapidities, and, unlike SC-ODT's, their distance does not grow with strain (supplementary material, § S3), which halves the shaded band of figure 21 and makes it strain-independent. A far-field discrimination of the mechanism requires a configuration whose acoustic response samples the sub-cap scales at accumulated strains of order unity—turbulence-ingesting rotors, where the streamtube contraction supplies the strain and the blade response the fine-scale weighting—and is left as the falsifiable target of the application study. I don't find any mention of such a future study. It should be added to Sec. 8.

Figure 22 assembles the two regimes on a single axis and summarises the section. In the laboratory window the frozen and computed inflows are indistinguishable and both are right: the window in which Amiet's theory has been validated is precisely the window in which its frozen-spectrum input costs nothing, which is why the practice has survived unchallenged. Beyond the window the frozen error grows monotonically—+0.6, +1.4, +3.4 dB in the band mean at $e = 0.5, 1, 2$ —with no measurement there to catch it, while the computed inflows remain within 1.1–2.7 dB of the reference and carry their measured uncertainty with them. What the methodology adds is the right-hand half of the figure: an inflow that continues to predict where the frozen assumption fails silently.

The scope of this statement is set out as carefully as its content. It concerns the *free* distortion by the mean strain: the surface-blocking effect, which measurements and simulations show *enhancing* the near-edge upwash at low frequency (Hunt & Graham 1978; dos Santos *et al.* 2023; Piccolo *et al.* 2024), is a distinct and partially compensating mechanism outside the present line model, and the strain history along individual streamlines is replaced by the canonical constant-rate distortion. What the calculation establishes is therefore not a noise

. This mechanism replaces [IS THIS CORRECT???

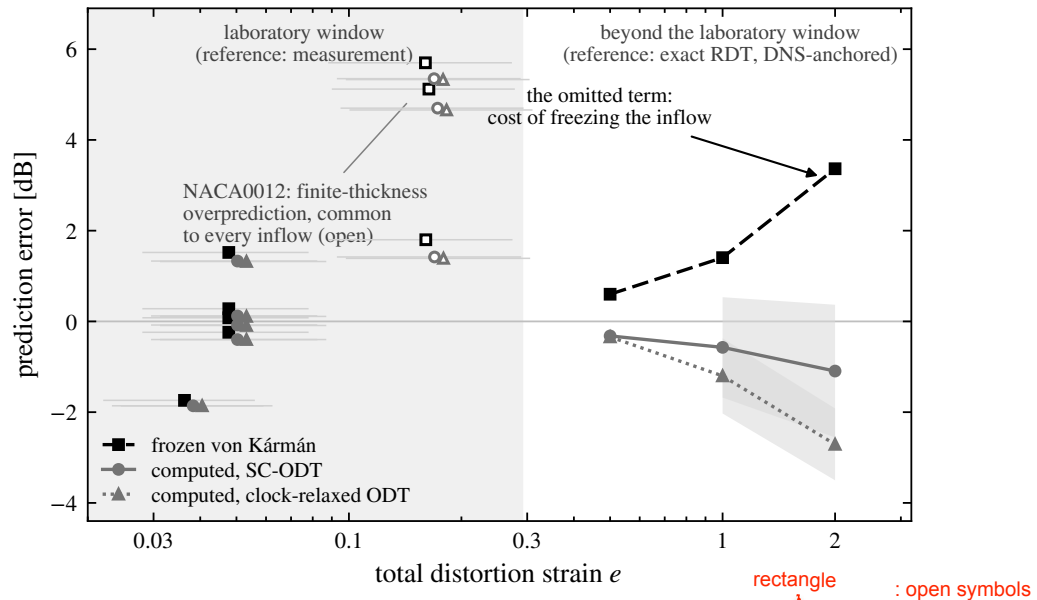


Figure 22. The cost of the frozen-inflow assumption against accumulated distortion: signed far-field prediction error, judged by the best available reference at each strain. Laboratory window (shaded): the eight measured cases of figure 20, bias over each case's comparison band—flat plates filled, NACA0012 open (its common overprediction is the finite-thickness effect); horizontal bars span the effective-strain truncation range, and the three inflow chains are offset slightly in ϵ for legibility. Beyond the window: band-mean error over 100–1000 Hz against the exact-RDT reference of figure 21 at $\epsilon = 0.5, 1$ and 2 ; shaded bands give the $\pm 10 \log_{10}(1 + \text{rms})$ spectral-distance uncertainty where measured ($\epsilon = 1, 2$).

How does one deduce this from looking at Fig. 19?

prediction but the magnitude and sign of the term that frozen-spectrum practice omits: order 5–20 dB at the energy-containing reduced frequencies for total strains of 0.5–2—far too large to neglect, and of the opposite sign to the blocking correction with which it must ultimately be combined. Figure 19 shows that the chain from model output to absolute far-field level is complete and cheap; what remains for a predictive comparison against measurement is the blocking correction and the streamline strain history, for which the present model supplies the distorted-inflow input at negligible cost, with its uncertainty quantified by the envelope of § 4.6.2 and its fidelity improved by the mechanism of § 6 precisely in the wavenumbers the response weights.

with greatest influence on

8. Conclusions

IS THERE ANYTHING WORTH MENTIONING HERE ABOUT THE CMK COMPARISON IN SEC. 4.5 AND ITS ELABORATION IN SEC. SM1?

We have presented a strain-coupled formulation of one-dimensional turbulence and, around it, a systematic account of what a single line of turbulence can and cannot say about rapid distortion. The formulation splits the distortion physics by structure: production enters as an exact continuous forcing, the rapid pressure-strain as an energy-conserving operator built from an explicit spectral closure, and the wavevector kinematics as a dilatation of the domain, while the eddy events of the standard model continue to carry the cascade and the slow return to isotropy. The construction preserves the model's structural invariants, reproduces the constant-free rapid-distortion onset rate identically, and is verified in the solver to the level of the closure it evolves (10^{-4} in the component fractions); at the moment level it reproduces the strained components of the Lee & Reynolds (1985) DNS quantitatively, with the one exception—the neutral-direction anisotropy—traced to the orientation information that every single-point closure discards, not to the line.

Three findings shape how such a model must be judged. First, the correct linear reference for line-projected spectra is broadband: exact rapid-distortion theory, projected onto a line, ^{rescales,} **redistributes** the projected upwash spectrum by a factor of 2.4 across the resolved range at $e = 1$, so a rigid spectral translation—the model’s own kinematics—is not the linear prediction, and comparisons built on it conflate a structural approximation with nonlinear physics. Second, against a strained-box DNS with exact-RDT companions reduced to the same observables, the strain-coupled model carries the correct integrated anisotropy but allocates it uniformly over wavenumber, ^{while} **where** linear theory and DNS concentrate it at the energy-containing scales; the deficiency is localised in the wavenumber-uniform rapid operator and grows with strain rapidity, a measured ^{artefact} **validity envelope** that accompanies the model into any application. Third, the single-line closure question admits an exact, falsifiable answer in place of a closure claim: line data bound the rapid term sharply, the bounds ^{valid} **are honest** up to $e \approx 0.25$ – 0.5 under plane strain and at all computed strains under axisymmetric distortion, and the axisymmetry hypothesis is measured rather than assumed—its plane-strain violation grows with accumulated strain and rapidity while the acoustically relevant upwash amplification remains robust to it.

The measured limits point to a specific missing ingredient, ^{that is thereupon identified and supplied} **which the paper identifies and supplies**. The strain-induced fine-scale anisotropy is delivered directly by large-eddy maps, which is why no acceptance-gating or map-reweighting scheme can control it; what does move it ^{supplementary} is relaxation applied at the scale of the imbalance—the model’s own energy-equalising kernel, applied hierarchically to eddy sub-scales. The intervention **family maps** ^{broadly reproduces the phenomenology} **the mechanism’s envelope precisely**: depth that reaches the measured scales is necessary and sufficient at moderate strain; dosage and delocalised application raise the early gains in proportion; and the fine-scale fade beyond $e \approx 3$, common to every scheme locked to the eddy clock, is removed by giving the relaxation its own clock—an independent stream of relaxation events whose size cap preserves the one-point physics. What remains is not timing but allocation: isotropisation is bounded at component equality and cannot reach the fine-scale sign reversal the DNS exhibits, so the remaining step is the wavenumber- and angle-dependent spectral kernel derived in § 5; exercising **it** in place of the wavenumber-uniform closure is the natural ^{future} continuation, now motivated by a measured deficit and pointed at a measured target.

For the application that motivated the work, the model delivers what the prediction ^{otherwise} chain lacks: a distorted, anisotropic, scale-resolved upwash spectrum, evolved rather than prescribed, at two orders of magnitude below the cost of a strained DNS ensemble—and § 7 carries it through Amiet’s theory to the radiated level. The assembled chain reproduces four published far-field experiments to 1–2 dB rms where the facilities are cleanest, with the computed correction causal—vanishing for thin plates, lowering the prediction towards the data for the NACA0012—and, against the linear reference at application strains, an evolved inflow ^{outperforms} **beats** the frozen one by a margin that grows with strain. Relative to the frozen-spectrum input of standard practice the computed distortion is a weaker leading-edge source at every resolved frequency—by 5–20 dB at the energy-containing scales for total strains 0.5–2, in the geometry-free kernel ratio and in the assembled absolute prediction at the reference configuration alike—because the distortion drains exactly the spanwise-coherent upwash content the aerofoil responds to. The omitted term is far too large to neglect and opposite ^{that} in sign to the surface-blocking correction ^{likewise include} **with which** a complete prediction must **combine it**; predictions inherit a spectral-shape uncertainty of ^{the} **order** the distance envelope at the operating point’s strain and rapidity, **stated here rather than discovered later**.

a falsifiable model prediction.

I find this summary of findings to be clearer than the characterizations of findings in Sec. 7. In Sec. 7, perhaps paraphrase these individual findings where applicable. Italics could be used to identify them as key takeaway messages.

Appendix A. The forward map and the bound kernels

This appendix derives the three ingredients that § 5 uses: the two-scalar representation of the axisymmetric solenoidal spectrum with its realisability cone, the forward map from the two scalars to the line spectra (with the falsifiers T1–T2), and the closure kernels (5.7)–(5.9), together with the linear programs that turn these linear relations into the sharp bounds of § 5.4.

A.1. The axisymmetric representation and realisability

The spectrum tensor of statistically homogeneous turbulence is Hermitian, positive semi-definite at each κ , and solenoidal, $\kappa_m \Phi_{mn}(\kappa) = 0$; it therefore has rank at most two, supported on the plane normal to κ . Under A1 (axisymmetry about $\mathbf{e}_2$, mirror-symmetric, helicity-free) it is fixed by two scalar functions of $(\kappa_2, \kappa_\perp)$ (Batchelor 1946; Chandrasekhar 1950; Lindborg 1995):

$$\Phi_{mn}(\kappa) = a(\kappa_2, \kappa_\perp) P_{mn}(\kappa) + c(\kappa_2, \kappa_\perp) \xi_m \xi_n, \quad \xi = \mathbf{e}_2 - \mu \frac{\kappa}{\kappa}, \quad \mu = \frac{\kappa_2}{\kappa}, \quad (\text{A } 1)$$

with $P_{mn} = \delta_{mn} - \kappa_m \kappa_n / \kappa^2$ the solenoidal projector, so that $\kappa_m \xi_m = 0$ and $|\xi|^2 = 1 - \mu^2 = 1 - x$ with $x = \kappa_2^2 / \kappa^2$: solenoidality is exact by construction, which is how the third would-be scalar of a general axisymmetric tensor is eliminated. For a unit vector $\mathbf{v}$ normal to κ , $\Phi_{mn} v_m v_n = a + c (\xi \cdot \mathbf{v})^2$, so the two eigenvalues of the polarisation matrix on that plane are a (attained for $\mathbf{v} \perp \xi$) and $a + (1 - x) c$ (for $\mathbf{v} \parallel \xi$). Positive semi-definiteness is therefore the pointwise *realisability cone*

$$a(\kappa_2, \kappa_\perp) \geq 0, \quad a(\kappa_2, \kappa_\perp) + (1 - x) c(\kappa_2, \kappa_\perp) \geq 0, \quad (\text{A } 2)$$

quoted in § 5.4. Isotropy is the special case $c = 0$ with a a function of κ alone.

A.2. The forward map to the line spectra

Writing $\kappa = (\kappa_\perp \cos \theta, \kappa_2, \kappa_\perp \sin \theta)$, the azimuthal moments at fixed $(\kappa_2, \kappa_\perp)$ are $\langle \kappa_1^2 \rangle_\theta = \langle \kappa_3^2 \rangle_\theta = \frac{1}{2} \kappa_\perp^2$, $\langle \kappa_1^4 \rangle_\theta = \langle \kappa_3^4 \rangle_\theta = \frac{3}{8} \kappa_\perp^4$, $\langle \kappa_1^2 \kappa_3^2 \rangle_\theta = \frac{1}{8} \kappa_\perp^4$, and every odd moment vanishes. Applied to (A 1) the diagonal weights average to

$$\langle P_{11} \rangle_\theta = \langle P_{33} \rangle_\theta = \frac{1}{2} (1 + x), \quad P_{22} = 1 - x, \quad \langle \xi_1^2 \rangle_\theta = \langle \xi_3^2 \rangle_\theta = \frac{1}{2} x (1 - x), \quad \xi_2^2 = (1 - x)^2, \quad (\text{A } 3)$$

and every mixed weight to zero. Substituting into the definition (5.3) of the line spectra, $\phi_{mn}(\kappa_2) = \int_0^\infty \langle \Phi_{mn} \rangle_\theta 2\pi \kappa_\perp d\kappa_\perp$, gives the **forward map** from the two scalars to the data a line carries:

$$\phi_{11}(\kappa_2) = \phi_{33}(\kappa_2) = 2\pi \int_0^\infty \left[a \frac{1+x}{2} + c \frac{x(1-x)}{2} \right] \kappa_\perp d\kappa_\perp, \quad (\text{A } 4)$$

$$\phi_{22}(\kappa_2) = 2\pi \int_0^\infty \left[a (1 - x) + c (1 - x)^2 \right] \kappa_\perp d\kappa_\perp, \quad (\text{A } 5)$$

$$\phi_{mn}(\kappa_2) = 0 \quad (m \neq n). \quad (\text{A } 6)$$

Equations (A 4)–(A 6) contain the two free falsifiers quoted in § 5.4: under A1 the two transverse line spectra coincide at every resolved wavenumber (T1), and the cross-spectra vanish (T2); both are properties the line itself reports, scale by scale, with no reference to unresolved data. The ϕ_{22} bracket is the same combination of a and c that appears in the kernel g_2 below—the conventions mesh, as they must, because ϕ_{22} and the 22-component of the rapid term weight the spectrum through the same projection geometry.

They also explain why the isotropic intuition about line data fails under A1. For isotropic turbulence ($c = 0$, $a = a(\kappa)$) the map is over-determined: one unknown function of one variable against two measured functions, with ϕ_{22} determining a by the classical Abel-type inversion and the transverse spectrum slaved to it by the derivative relation. Under A1 the bookkeeping reverses: *two unknown functions of two variables* against two measured functions of one variable. At each fixed κ_2 the data supply exactly two numbers—the two weighted $\kappa_\perp$ -integrals (A 4)–(A 5)—about two unknown functions of $\kappa_\perp$; any perturbation $(\delta a, \delta c)(\kappa_\perp)$ annihilating both integrals while respecting (A 2) is invisible to the line. This infinite-dimensional null space is the quantitative content of the obstruction of § 5.2 after the axisymmetric collapse.

A.3. Derivation of the closure kernels

For the diagonal components the exact result (5.2) reads

$$\Pi_{nn}^{(r)} = 2 A_{kl} \int_{\mathbb{R}^3} \frac{\kappa_n \kappa_k}{\kappa^2} \Phi_{nl}(\boldsymbol{\kappa}) d\boldsymbol{\kappa} \quad (\text{no sum on } n). \quad (\text{A } 7)$$

Substituting (A 1), carrying out the azimuthal average with the moments above, and specialising to $A = \text{diag}(\frac{1}{2}, -\frac{1}{2}, 0)$ reduces each component to a $2\pi\kappa_\perp$ -weighted integral over the half-plane of a polynomial combination in x , linear in (a, c) —the kernels (5.7)–(5.9) of the main text. The upwash component illustrates the mechanics: only $k = l = 1$ and $k = l = 2$ contribute, with azimuthal averages $\langle \kappa_1 \Phi_{21} \rangle_\theta = -\frac{1}{2} \kappa_\perp^2 \kappa_2 [a + (1-x)c]/\kappa^2$ and $\langle \Phi_{22} \rangle_\theta = (1-x)[a + (1-x)c]$, so that both enter through the same bracket and

$$\langle \text{integrand of (A 7)} \rangle_\theta \Big|_{n=2} = 2 \left[-\frac{1}{2} \cdot \frac{x(1-x)}{2} - \frac{1}{2} x(1-x) \right] [a + (1-x)c] = -\frac{3}{2} x(1-x) [a + (1-x)c], \quad (\text{A } 8)$$

which is $g_2(x)$. The transverse components involve in addition the fourth moments of (A 3)’s underlying table and yield g_1 and g_3 ; the full reduction has been verified symbolically against (A 7). Three identities check the algebra: $g_1 + g_2 + g_3 = 0$ at every x for both scalars (pointwise energy conservation); each kernel vanishes at $x = 1$, where the wavevector aligns with the line and the projection degenerates; and averaging over the isotropic angular measure $d\mu$ on $[0, 1]$ gives $\langle g_1 \rangle : \langle g_2 \rangle : \langle g_3 \rangle = +\frac{1}{5} : -\frac{1}{5} : 0$, reproducing $\Pi_{ij}^{(r)} = \frac{4}{5} k_t S_{ij}$ (Crow 1968) as in § 5.7.

A.4. The bounds as linear programs

The target (5.6), the data constraints (A 4)–(A 5), and the cone (A 2) are all linear in (a, c) , and everything decomposes over the resolved wavenumber: writing $\Pi_{nn}^{(r)} = \int_0^\infty \pi_n(\kappa_2) d\kappa_2$ with $\pi_n = 2\pi \int_0^\infty g_n(x) [a, c] \kappa_\perp d\kappa_\perp$, the extremisation over all spectra consistent with the line data is a family of *independent* infinite-dimensional linear programs, one per κ_2 :

$$\pi_n^\pm(\kappa_2) = \max / \min_{(a,c)(\kappa_\perp)} 2\pi \int_0^\infty g_n(x) [a, c] \kappa_\perp d\kappa_\perp \quad \text{s.t.} \quad (\text{A 4})\text{--}(\text{A 5}) \text{ hold, (A 2) pointwise,} \quad (\text{A } 9)$$

whence $\Pi_{nn}^\pm = \int \pi_n^\pm d\kappa_2$ and the indeterminacy $\mathcal{I}_n = (\Pi_{nn}^+ - \Pi_{nn}^-)/2k_t S$ of § 5.4, normalised by the natural rapid-term scale so that $\mathcal{I}_n$ is comparable across strains (the isotropic value is $\Pi_{11}^{(r)} = \frac{2}{5} k_t$ per unit $S_{11} = \frac{1}{2}$). Discretised on a logarithmic $\kappa_\perp$ grid of order 10^2 points, (A 9) is a linear program with two equality constraints and the cone, solved per κ_2 in milliseconds; the bounds are global and sharp by LP duality—no fitting and no local minima. The side conditions of § 5.4 enter as further linear constraints: the log-Lipschitz slope cap $|d \ln(a, c)/d \ln \kappa_\perp| \leq \lambda$ and the polarisation cap $|c| \leq \gamma a$. The LP extremals concentrate on few-point supports in $\kappa_\perp$ (bang–bang spectra), which is why the raw bounds are conservative and why smoothness alone barely narrows them: a log-Lipschitz spectrum can still vary like $\kappa_\perp^{\pm\lambda}$, so the angular reweighting that drives the indeterminacy survives smoothing, while the polarisation cap attacks it directly. The dual variables of (A 9) certify, per resolved wavenumber, which linear combinations of the measured line spectra control each π_n ; finite-window bias on ϕ_{mn} at small κ_2 propagates linearly and can be carried as interval data in place of the equalities with no change of machinery.

Appendix B. Numerical parameters, initial conditions and statistics

Referenced only on line 853 in a context unrelated to numerical implementation

B.1. ODT configurations

All line ensembles share the solver settings of the standard model (Stephens & Lignell 2021): eddy-rate constant $C = 7$, viscous-penalty parameter $Z = 450$, kernel parameter $\alpha = 2/3$, adaptive mesh with $dx \in [2 \times 10^{-4}, 2 \times 10^{-2}]$ on a unit domain (4000 initial cells), explicit advancement with diffusive CFL 0.5 and a strain-resolved bound of 10^{-2} on the total-strain increment per step (§ 3.3). The initial condition is the resolved spectral field of § 3.3: a band-limited isotropic superposition of 64 Fourier modes peaked at eight wavelengths per domain, which places no energy at the grid scale. Strained production ensembles use an unstrained precursor: the line evolves as standard ODT until the eddy population is statistically active and the initialisation transient has relaxed, and the strain is switched on only then (§ 4.6). The molecular viscosity is set per study: 10^{-4} for the transmission diagnostic of § 6 (a scale diagnostic, resolved with wide margin), 10^{-5} for the strain-on/off ensembles of § 4.3, and 3×10^{-6} on a 16000-cell grid at the operating point of § 4.4.